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Effect of high-temperature wettability of glass on interfacial contact quality during metallization of silver electrodes in solar cells

Qian Li [a], Wenbin Sun [a], Yinghu Sun [a], Xiaojie Liu [a], Hanying Wang [c], Shenghua Ma [b]

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Highlights

- The addition of Bi_2O_3 boosts the dissolution and reduction rates of silver ions, decreases glass surface tension, and prevents the aggregation and movement of silver particles.

- Good wettability enables the glass phase to cover the silicon emitter surface uniformly, promoting the creation and even spread of nano-silver colloids at the interface.

- Proper flow and spreading of molten glass help fill voids formed during silver paste sintering, resulting in improved electrical conductivity of the paste.

- Adequate glass wettability contributes to reliable silver-silicon contact, reduces series resistance and enhances fill factor in solar cells.

Abstract

The efficiency of tunnel oxide passivated contact (TOPCon) solar cells depends on achieving high-quality metallization, with glass frits playing a critical role in promoting effective metallization and establishing stable ohmic contact at the Ag-Si interface. This study primarily investigated the flowability and wettability of different glass frits at high temperatures by adjusting the composition of low-lead glass, and analyzed their impact on the contact quality and electronic transport properties at the silver-silicon interface. The results demonstrated that excessive addition of Bi_2O_3 can lead to over-wetting of the glass frit. Overly etching of glass frit could pose a risk of damaging the silicon substrate. Inadequate wettability can lead to uneven contact interfaces after sintering, which degrades the silver-silicon contact performance. However, appropriate adjustment of the content of Bi_2O_3 and TeO_2 can lower the glass transition temperature (T_g) of the glass frit, promoted the dissolution and flow of silver powder. It also improved the wettability of the glass frit on the substrate material at high temperatures, facilitating the formation of uniform and dense electrode grid. Moreover, it can prevented excessive aggregation of silver particles, promoted the formation of uniformly distributed nano-silver crystallites on the silicon emitter surface. This improved the silver-silicon contact quality, enable more efficient current transfer from the silicon substrate to the electrode. And ultimately improves the photoelectric conversion efficiency of the solar cell.

Previous Next

Keywords

Glass frit; High-temperature wettability; Ag-Si contact; Metallization

1. Introduction

Crystalline silicon solar cells continue to dominate the photovoltaic market due to their technological maturity, stable material supply, high electrical performance, and adaptable industrial production processes [1,2]. Among them, TOPCon cells stand out for their high-efficiency passivated contacts and selective carrier transport. This structure allows for higher photovoltaic efficiency, lower series resistance (R_S), and sustained long-term stability [3,4]. The n-type TOPCon solar cells are increasingly replacing p-type cells, becoming the mainstream choice due to their higher efficiency, low degradation rates, excellent temperature characteristics, and compatibility with existing production processes [5]. Furthermore, with advancements in technology and economies of scale, the production costs of n-type TOPCon cells are decreasing, making them a key player in the photovoltaic market [6,7].

In the structure of TOPCon cells, the delicate balance between the surface passivation layer and the metallization layer is essential [8,9]. The primary function of the passivation layer is to reduce surface recombination, thereby increasing minority carrier lifetime and open-circuit voltage (V_{oc}). However, efficient current collection depends on good ohmic contact that connects the metal electrode with the Si wafer [10,11]. If the passivation layer is damaged, surface recombination rates increase, causing more electrons and holes to recombine in the contact area, thereby reducing the minority carrier lifetime. As n-type TOPCon technology progresses, efficiency continues to improve. One of the key factors driving this progress is the optimization of metallization processes. Metallization is critical for solar cells as it directly affects their conductivity, efficiency, and stability [12,13]. However, among the various metallization techniques, achieving high-quality electrical contacts without compromising the integrity of this complex structure remains a significant challenge [14].

It has been reported that the laser-enhanced contact optimization (LECO) improves the sintering and contact between the silver paste and the solar cell by applying localized laser heating [15,16]. The central role of the LECO treatment is to optimize the interfacial contact between the metal electrode and the silicon wafer by means of laser technology. By precisely heating the silver paste with a laser, the passivation layer is locally disrupted, allowing silver to directly contact the silicon. This forms a low-resistance metal-silicon contact point, significantly reducing contact resistance. Glass frit, an essential component of the silver paste, directly affects the metallization and overall performance of the solar cells. During silver paste sintering, the melting glass coats silver particles and fills the gaps. Once fully melted, the glass frit creates a continuous liquid phase at the interface. At this point, good wettability ensures the even distribution of metal particles across the silicon surface, promoting effective contact between the silver paste and the wafer, thereby reducing interfacial resistance and improving ohmic contact [17,18]. Thus, the uniformity of contact interface layer is largely determined by the wettability of the glass frit. Moreover, chemical reactions between the frit and the silicon surface enhance adhesion, preventing delamination or oxidation at the contact interface. In conclusion, glass frit plays several critical roles during the solar cell sintering process [19,20].

In this study, under the conditions of LECO treatment, we modified the composition of the glass frit to obtain samples with distinct softening points and wettability. Scanning electron microscopy was utilized to observe etched silicon wafers, focusing on the flow and uniform spreading of glass frits during sintering. It further analyzes the impact of glass frits with varying wetting properties on the Ag-Si contact interface and the deposition of silver crystals on the n^+ emitter surface. This allows us to explore the close relationship between the wetting behavior of different glass frits and the electrical performance of the cell.

2. Experimental

2.1. Preparation of glass frits

The reagents used in this experiment were purchased from China Pharmaceutical Group Chemical Reagent Co., Ltd. The reagents were utilized directly, without any additional purification. The components of the glass frit were detailed in Table 1. A conventional melting and quenching method was used to prepare the glass frits. The oxides were carefully weighed, thoroughly mixed in an agate mortar. And subsequently calcined in a muffle furnace. Subsequently, the glass was quenched in deionized water, dried and ball-milled to the desired target particle size (1–3 μm).

Table 1. Experimental compositions of glass frits (wt%).

Glass frit	PbO	SiO_2	ZnO	B_2O_3	TeO_2	Bi_2O_3	Others
L1	40.8	6	10	20	10	0	13.2
L2	41.5	6	10	20	7.3	2.8	12.4
L3	41.5	6	10	20	5	5.2	12.3
L4	40.2	6	10	15	10	13	5.8

2.2. Preparation of silver paste and screen printing

The formulation of silver paste involved blending silver powder, an organic binder, and various glass frits in the proportion of 88.0:2.5:9.5. This mixture was processed through a three-roll mill at 500 rpm for three passes to achieve uniform dispersion. Zhongwei New Material Co., Ltd. provided the silver powder and organic binder for the silver paste. The resulting silver pastes derived from glass frits L1, L2, L3, and L4 were labeled as LA-1, LA-2, LA-3, and LA-4, respectively. A screen printing machine was employed to print the silver paste onto the p+emitter of the n-type monocrystalline silicon wafer (182 mm × 182 mm), and then sintered in a belt furnace with a peak temperature of 730 °C. After sintering, all solar cells underwent LECO treatment, where a focused laser beam laterally scanned the surface to locally activate carriers. Depending on the difference in silver paste, the solar cells were distinguished and designated LC-1, LC-2, LC-3 and LC-4. At the same time, the standard cells are prepared in exactly the same way. The schematic diagram of silver grid screen printing and laser processing for crystalline silicon solar cells was displayed in Fig. 1.

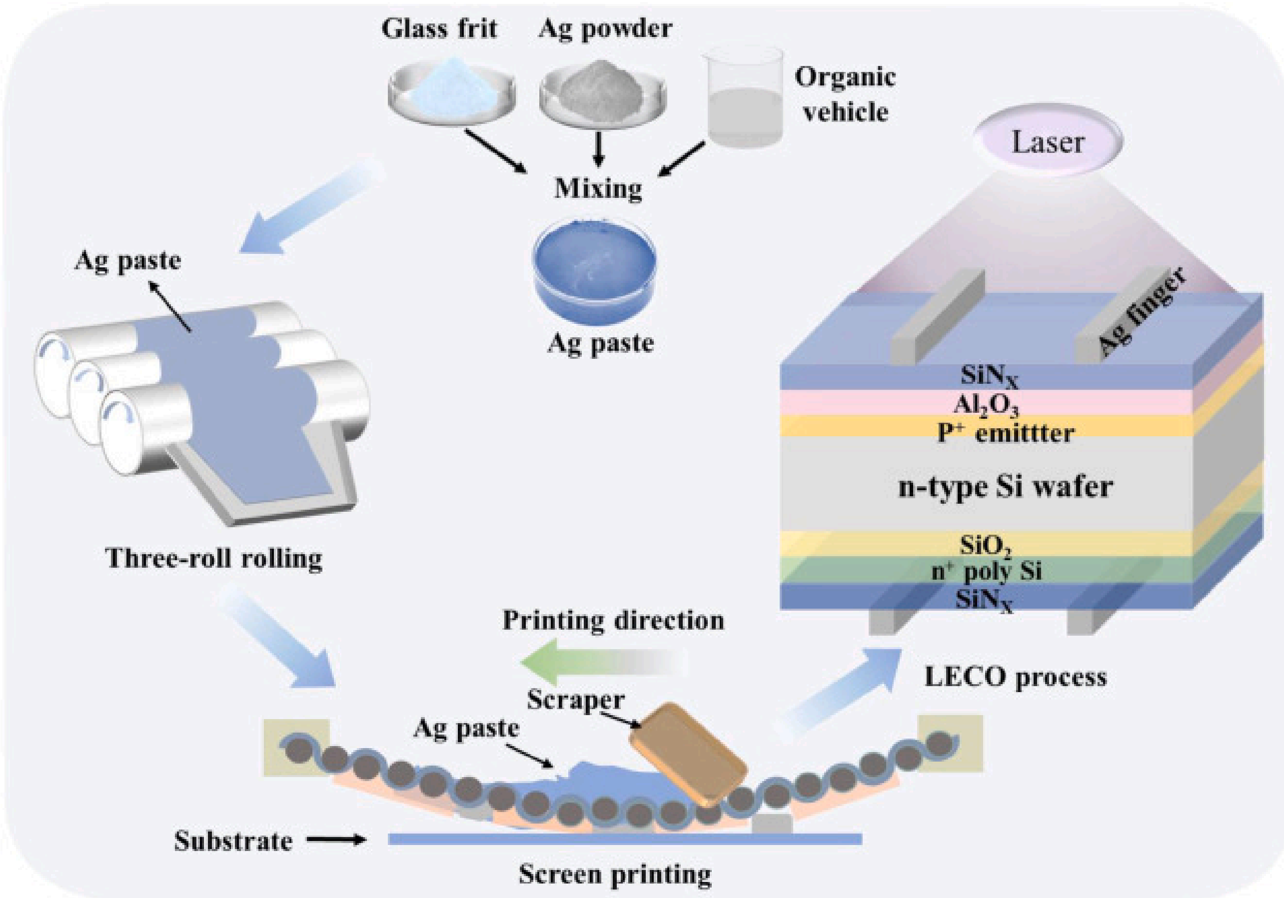

Fig. 1. The schematic diagram of silver finger printing and laser processing for crystalline silicon solar cells. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

2.3. Characterizations

The amorphous structure of the glass frits was studied by X-ray diffraction (XRD, Bruker D8 Advanced, Germany). The glass frits morphology and the cross-sectional microstructure of the cell were examined by scanning electron microscopy (SEM, Zeiss σ300, Germany). The microstructure of the glass layer and emitter surface was further investigated after sequential etching of the silicon wafer. The energy dispersive spectroscopy (EDS) was used to analyze the elemental composition of diffusion and spreading in silver pastes. The Fourier transform infrared spectroscopy (FT-IR, Bruker Tensor27) facilitated the analysis of peak positions in the glass frits and assisted in assessing alterations in chemical bonds and functional groups. The T_g was tested using a differential scanning calorimeter (DSC, NETZSCH STA 449C, Germany) at a heating rate of 10 °C/min. The chemical composition and valence states of elements of the glass frits were examined by X-ray photoelectron spectroscopy (XPS, PHI5000, VersaProbeIII, Japan). The change in the contact angle of glass with the silicon substrate upon high-temperature melting was determined with a contact angle goniometer (KRUSS, DSA25, Germany). A 3D digital microscope (Zeta-20, USA) was used to observe the surface and aspect ratio of the silver grid. The TOPCon solar cells were tested for their photovoltaic performance using an EKOI-V Tracer under standard test conditions (STC: 1000 W m^{-2} of solar radiation, 25 °C).

3. Results and discussion

Fig. 2(a) displays the XRD patterns for the four varieties of glass frits. The patterns were very similar, with a pronounced scattering characteristic typical of amorphous materials. Aside from a broad peak around 28°, no distinct crystalline peaks were observed, confirming that the produced glass frit samples exhibit a well-defined amorphous structure. In Fig. 1(b), the SEM image reveals the microstructure of the glass frits, indicating particle sizes between approximately 1-3 μm. In Fig. 1(c)–(j), the SEM elemental mapping displayed Si, Al, Pb, Te, Bi, Zn, and O in the glass material, indicating the homogeneity of the distribution.

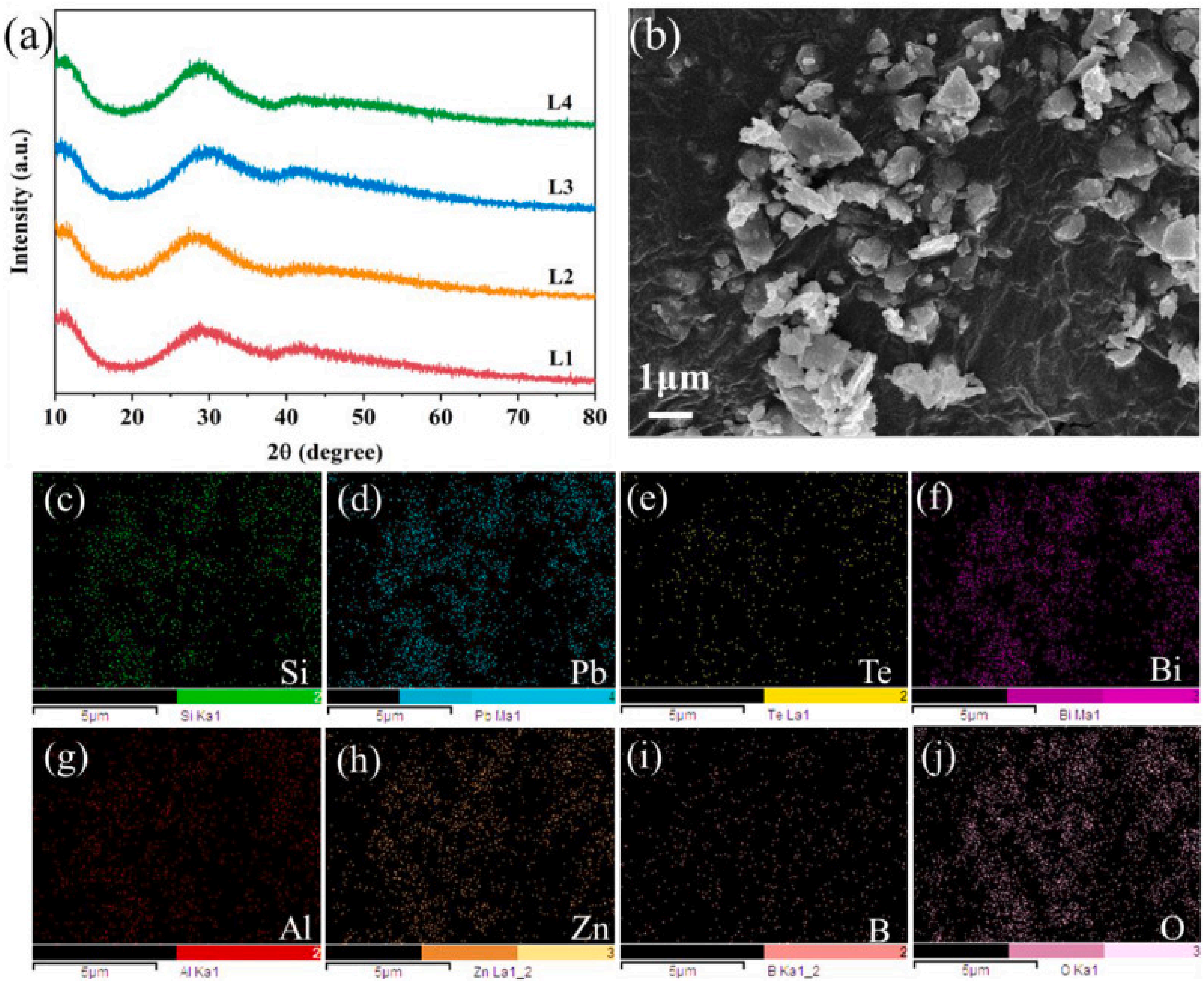

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Fig. 2. (a) The XRD spectrum of glass frit samples; (b) The SEM micromorphology of L3 glass frit; (c)–(j) The elemental mappings of the L3 glass frit.

As depicted in Fig. 3(a), the peak positions for all four glass frits were largely similar. L2, L3 and L4 glass frits contain Bi_2O_3, the peaks near 510 cm^{-1} were attributed to the bending vibrations of the Bi-O-Bi bonds in the $[BiO_6]$ octahedra [21,22], and the peaks were not pronounced because of the lesser addition of L2. The absorption peak at 700-710 cm^{-1} corresponds to the symmetric vibrations of Te-O bonds within the $[TeO_3]$ trigonal pyramids [23,24]. The peak around 1000 cm^{-1} arose from the symmetric stretching vibrations of Si-O bonds in the $[SiO_4]$ tetrahedra [25], indicating that the glass network structure is primarily composed of Te-O and Si-O bonds. The peak near 1340 cm^{-1} was due to the asymmetric stretching vibrations of B-O bonds in the $[BO_3]$ units [26,27]. The peak around 1640 cm^{-1} results from the bending vibrations of O-H bonds in water molecules, while the peak at 2354 cm^{-1} arose from water vapor generated during the KBr pellet preparation. In addition, the peak near 3445 cm^{-1} was attributed to the symmetric stretching of O-H bonds [[28], [29], [30]].

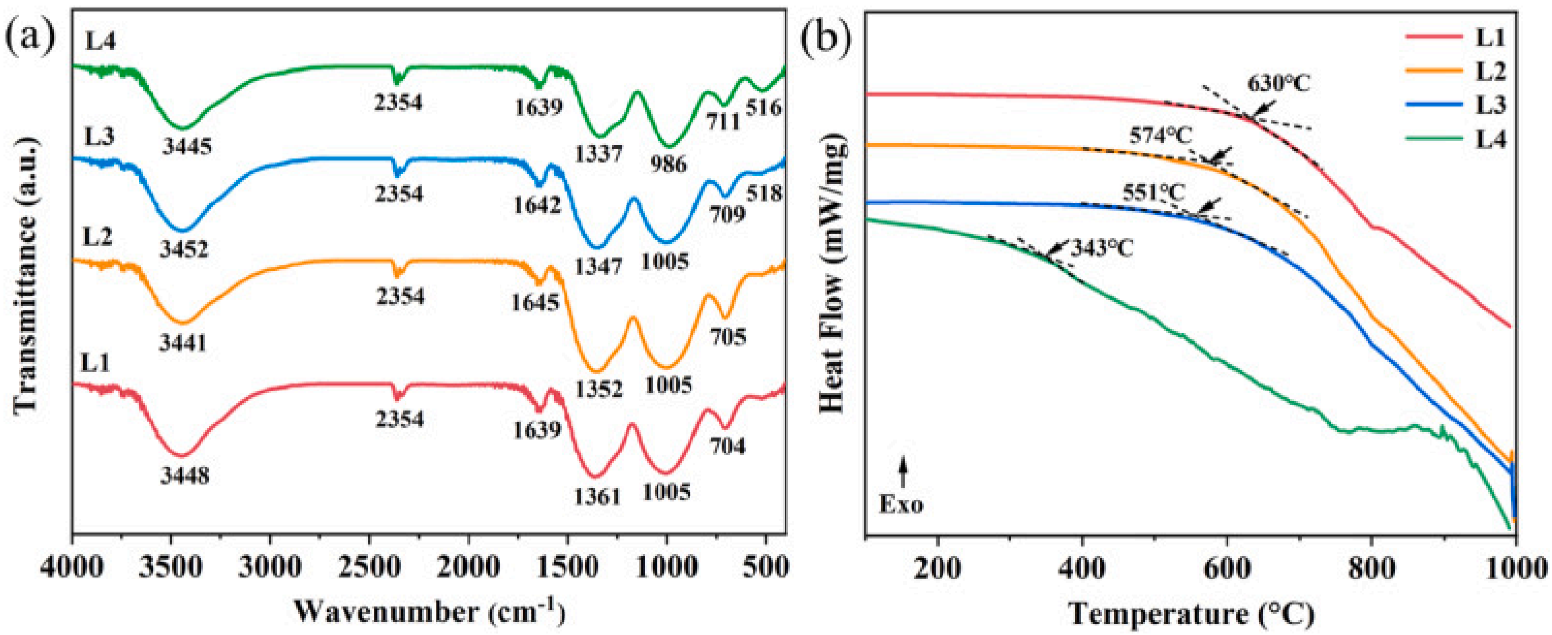

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Fig. 3. (a)The FT-IR spectrums of the glass frits; (b) The DSC curves of the glass frits.

The T_g of the glass frits determines their melting behavior during the sintering process, which in turn affects both the Ag-Si contact, as well as the overall electrical performance of the cells. The heat flow curves of different glass frits were illustrated in Fig. 3(b). The L1 glass frit, which contains no Bi_2O_3, exhibits a relatively high T_g. As the Bi_2O_3 content in the glass frit increases, the T_g of the glass frit gradually decreases. Notably, the L4 glass frit, with the highest Bi_2O_3 content, has the lowest T_g at 343 °C. While this was likely due to Bi_2O_3 acting as a network intermediate, possessing significant polarization capability. The high charge and larger size of Bi^{3+} ions can polarize surrounding oxygen atoms, weakening the bonding forces between oxygen atoms and other network formers, such as Si^{4+} [31]. The bond energy of Bi-O (337.2 kJmol^{-1}) was much weaker than that of Si-O (460 kJmol^{-1}), facilitated atomic and molecular mobility. This leads to a more loosely structured glass network, thereby reducing the softening temperature. Nevertheless, a lower T_g is not always advantageous. If the glass frits have an excessively low transition temperature, they began to dissolve too early during the sintering process and exhibited high fluidity. This accelerates sintering, and driven by surface energy reduction, the dissolved silver particles tend to agglomerate. This results in dense regions in some areas, while other areas form voids due to the movement of molten silver. This compromises the density and ultimately increasing the resistance of silver grids.

The characterization method of X-ray photoelectron spectroscopy (XPS) was utilized to meticulously analyze the chemical composition and the valent states of the constituents within the glass frits. Fig. 4(a) presented the comprehensive spectrum of the glass frits. In Fig. 4(b), the binding energy associated with the C-C bond is observed at 284.80 eV, while the O-C=O bond exhibited a binding energy of 287.19 eV. Fig. 4(c) and (d) present the XPS spectra of B and Si, respectively, both exhibiting single-peak characteristics with binding energies of 192.21 eV and 101.91 eV. The XPS spectrum of Al (Fig. 4(e)) displays a characteristic peak at 74.48 eV, which aligns with the binding energy of aluminum in an aluminosilicate configuration. In vitreous networks, aluminosilicates typically act as network intermediates through tetrahedral coordination of Al^{3+} [32,33]. This structural integration enables the formation of highly cross-linked glass networks stabilized by strong covalent Al-O-Si bonds, thereby enhancing both thermal stability and chemical durability. In Fig. 4(f), the Pb spectrum exhibits two distinct peaks at binding energies of 138.64 eV and 143.47 eV, which can be attributed to the presence of Pb^{2+} [34], corresponding to the Pb $4f_{7}/_{2}$ and Pb $4f_{5}/_{2}$ spin-orbit doublet, respectively. Fig. 4(g) presents the Te 3d spectrum, where the two peaks observed at binding energies of 577.73 eV and 588.12 eV, which may exist in the glass as [TeO_4] structural units [35]. The XPS spectrum of Bi reveals binding energies of 159.30 eV for Bi $4f_{7/2}$ and 164.56 eV for Bi $4f_{5/2}$, indicating the presence of Bi^{3+} (Fig. 4(h)) [36]. Lastly, Fig. 4(i) exhibits the spectrum of Zn with Zn $2p_{3/2}$ and Zn $2p_{1/2}$ peaks observed at 1021.98 eV and 1045.08 eV (Δ=23.1 eV). The chemical shifts of binding energy and spin-orbit splitting energies indicate that Zn^{2+} was presented in the glass frits [37]. The elements in the glass are present as stable oxides.

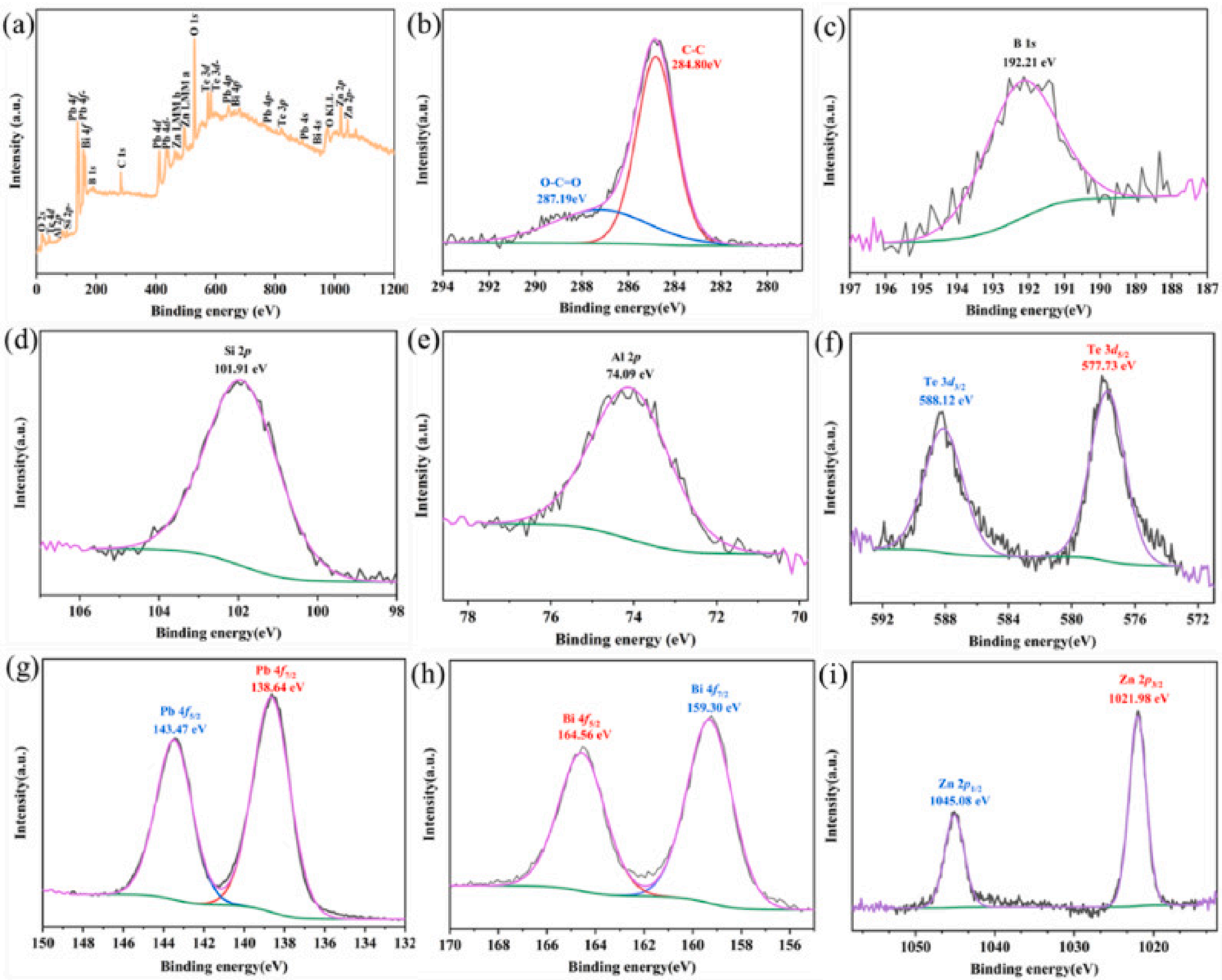

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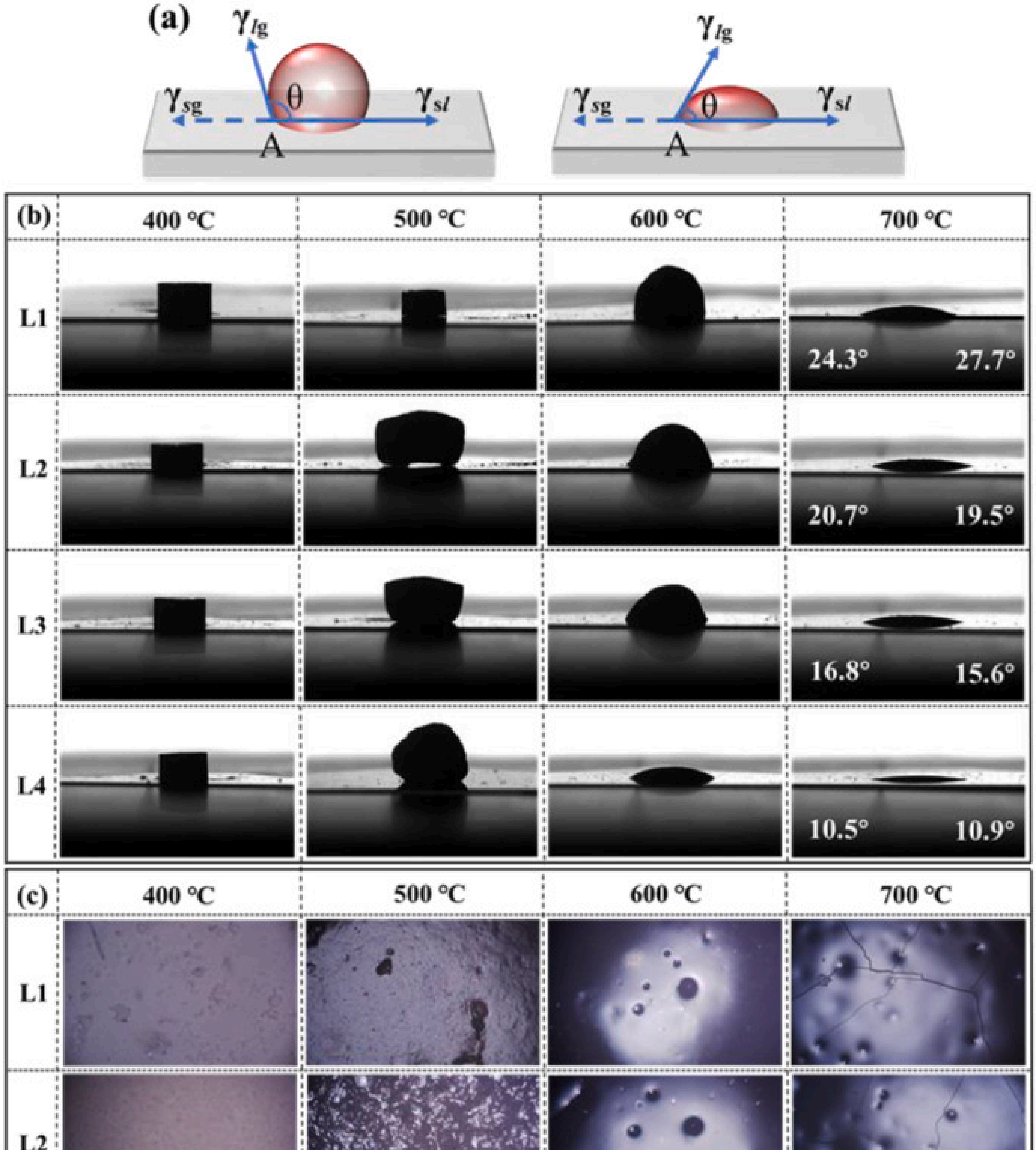
Fig. 4. (a) The XPS spectra of the L3 glass frit; (b)–(i) The XPS spectra of each element comprising L3 glass frit.

The contact angle is a measure of the wettability of a surface by a liquid. According to Young's equation (Equation (1)), the contact angle θ is determined by the interfacial tensions at the solid-gas (γ_{sg}), solid-liquid (γ_{sl}), and liquid-gas (γ_{lg}) interfaces [38,39]. Fig. 5(a) depicted the contact angle model between the glass frit and the silicon substrate. A contact angle below 90° denotes good wettability, which allows the glass frit to unfold on the silicon substrate evenly. However, a contact angle greater than 90° suggests poor wettability, which may lead to poor contact and a decline in cell performance [40].

$$\cos\ \theta = \frac{\gamma_{sl-}\gamma_{sg}}{\gamma_{lg}} \tag{1}$$


(a)
γlg
γsg
θ
γsl
A
(b)
400 °C
500 °C
600 °C
700 °C
L1
24.3°
27.7°
L2
20.7°
19.5°
L3
16.8°
15.6°
L4
10.5°
10.9°
(c)
400 °C
500 °C
600 °C
700 °C
L1
L2

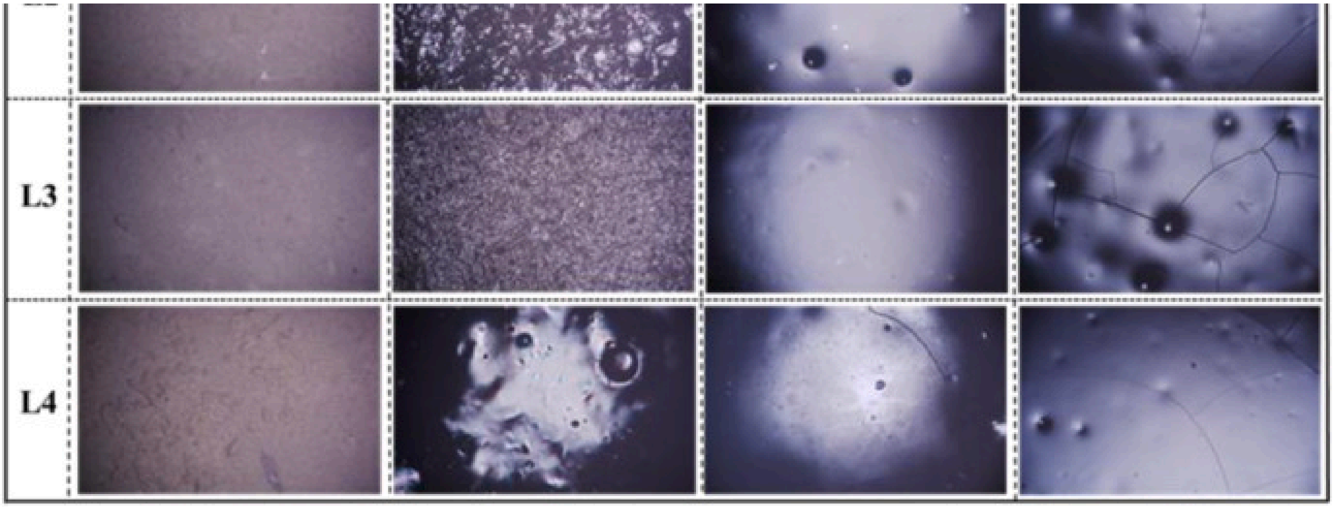


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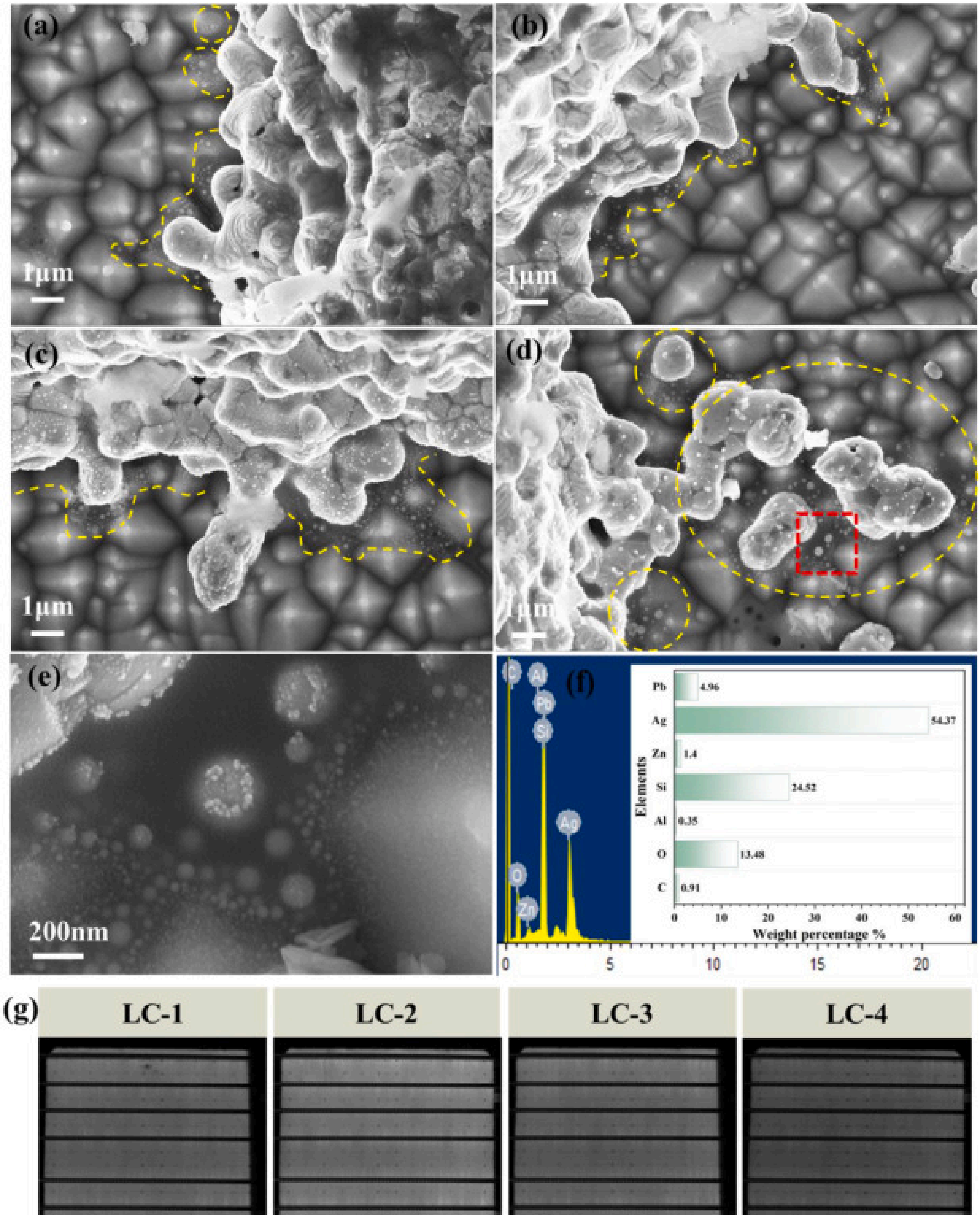
Fig. 5. (a) Schematic representation of the contact angle between the molten glass and the substrate. (b) Contour view of glass frits after sintering tested with a contact angle meter. (c) The morphological transformations of glass frits at various sintering temperatures.

Fig. 5(b) illustrates the melting state of different glass frits sintered for 1 min at various temperatures, analyzed using a contact angle tester. The 3D microscope images in Fig. 5(c) show the surface morphology after sintering. Before testing, the glass frit was pressed into cylindrical shapes with a 3 mm diameter base. At 400 °C, the glass frit samples retained their cylindrical shape, indicating no melting. Small surface protrusions were observed on the L2 and L3 samples, while L1 and L4 glass frit showed some visible unmelted particles on their surfaces. Significant changes were observed starting at 500 °C. L1 glass frit maintained its cylindrical shape but exhibited volume reduction and surface roughness. L2 and L3 glass frits expanded into a trapezoidal shape, with L3 glass frit showing a relatively smooth surface. On the other hand, L4 glass frit took on a rough semicircular form with a mirror-like surface. The limited flow and inconsistent wetting behavior of L1, L2, and L4 glass frits during melting led to uneven surface textures. At 600 °C, the glass samples formed hemispherical shapes, with L4 glass frit displayed a significantly higher axial shrinkage rate than radial shrinkage. This is attributed to the strong oxygen affinity of heavy metal ions (Bi^{3+} and Te^{4+}) in Bi_2O_3 and TeO_2, which chemically bond with the silicon surface, reducing interfacial energy and enhancing wetting and spreading [41,42]. Besides that, the L1 and L2 glass frits showed surface porosity, likely due to gas being trapped in the incompletely melted glass. In contrast, the L3 glass frit appeared relatively smooth. L4 glass frit exhibited excessive wetting and high-temperature flow on the silicon substrate, which could lead to uneven thermal stress distribution and crack formation during cooling. At 700 °C, all samples exhibited surface tension, with varying degrees of cracks and pores. In summary, the L3 glass frit, containing small amounts of bismuth and tellurium, demonstrated better flow characteristics and more suitable wetting on the silicon substrate.

Fig. 6(a–e) show partial surface morphologies of the silver grids. The flowing and spreading phenomenon of the silver paste is clearly visible in the yellow regions. The EDS analysis indicates that the extended area consists of both glass melt and silver crystals. In the case of the L4 cell, the silver paste spreads beyond the edge of the grid line after printing, leading to an increase in the width of the electrode. The high polarization of Bi_2O_3 and TeO_2 in the glass reduces the surface tension of the glass melt, thereby enhancing its spreading ability [43]. The sharp decrease in shunt resistance (R_{sh}) and the drop in V_{oc} for the LC-4 cell indicate severe leakage current caused by internal and interfacial defects. Additionally, the EL image darkens, showing a significant increase in interface recombination. Therefore, excessive wetting reduces the viscosity of the glass phase, promotes interfacial reactions, and facilitates the migration of silver particles. However, this can lead to the aggregation of silver particles in localized areas, increasing the risk of silver penetrating the passivation layer and forming local short-circuit points. The LC-2 cell has the highest R_{sh}, and the EL image is relatively brighter, indicating a decrease in leakage current and less recombination loss [43]. The L2 and L3 glass frits exhibit moderate wetting properties, resulting in denser silver grids in the cells. Bi^{3+} ions, with their large ionic radii and low coordination numbers, tend to occupy gaps within the glass. Particularly in areas with insufficient contact, silver particles redistribute through flow and spreading during sintering. Moderate flow and spreading help fill microscopic gaps between the silver grids and the silicon substrate, facilitating the densification of the silver film. This enhances the formation of a dense conductive network, boosting current transport efficiency and the conductivity of the silver paste.


(a)
1μm
(b)
1μm
(c)
1μm
(d)
1μm
(e)
200nm
(f)
C
Al
Pb
Si
Ag
O
Zn
Pb 4.96
Ag 54.37
Zn 1.4
Si 24.52
Al 0.35
O 13.48
C 0.91
Elements
Weight percentage %
0 10 20 30 40 50 60
0 5 10 15 20
(g)
LC-1
LC-2
LC-3
LC-4

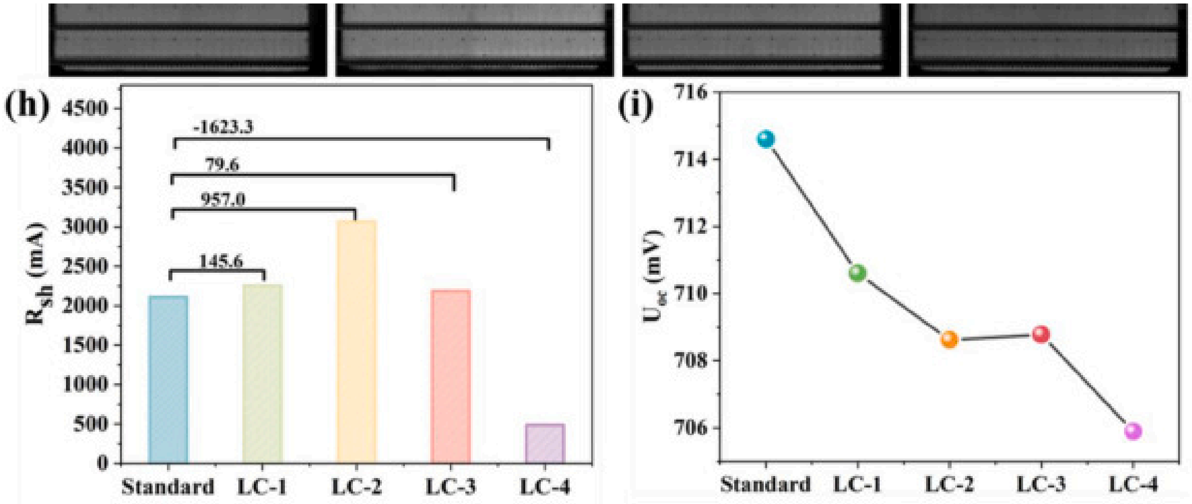


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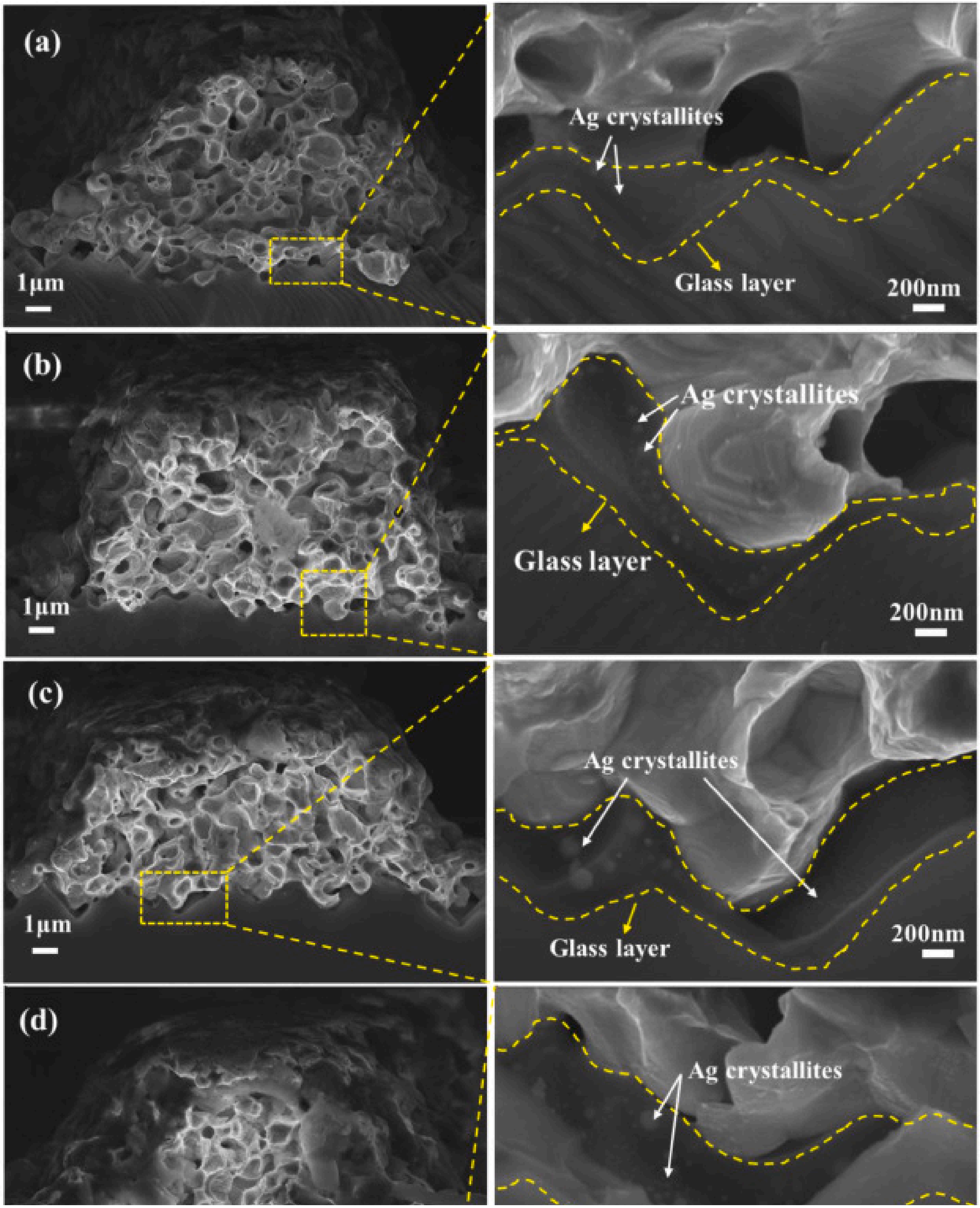
Fig. 6. Surface topography of the edge of the silver grids: (a) LC-1, (b) LC-2, (c) LC-3, (d) LC-4; (e) Enlarged view of the red area; (f) Element distribution maps of the red area, (g) EL images of different TOPCon solar cells, (h–i) Comparison of R_{sh} and V_{oc} of TOPCon solar cells with standard cells. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

The schematic diagram of the glass frit forming a contact between the silver electrode and the silicon wafer is shown in Fig. 7(e). During the sintering process, the glass powder softens and wets the Ag powder flowing to the Si substrate. Ag reacts with oxygen to dissolve in the glass layer, and part of Ag^{+}reacts with SiN_x and Si substrate. At last, irregularly shaped silver crystallites were formed on the Si surface, as described in equations (2), (3), (4) [44,45].

$$4Ag_{(s)}+O_{2(g)} \rightarrow 2Ag_2O_{(in\ glass)} \tag{2}$$

$$2Ag_2O_{(in\ glass)}+SiNx_{(s)} \rightarrow 4Ag_{(s)}+SiO_{2(in\ glass)}+x/2N_{2(g)} \tag{3}$$

$$2Ag_2O_{(in\ glass)}+Si_{(s)} \rightarrow 4Ag_{(g)}+SiO_{2(in\ glass)} \tag{4}$$



(a)
1µm
Ag crystallites
Glass layer
200nm
(b)
1µm
Ag crystallites
Glass layer
200nm
(c)
1µm
Ag crystallites
Glass layer
200nm
(d)
Ag crystallites

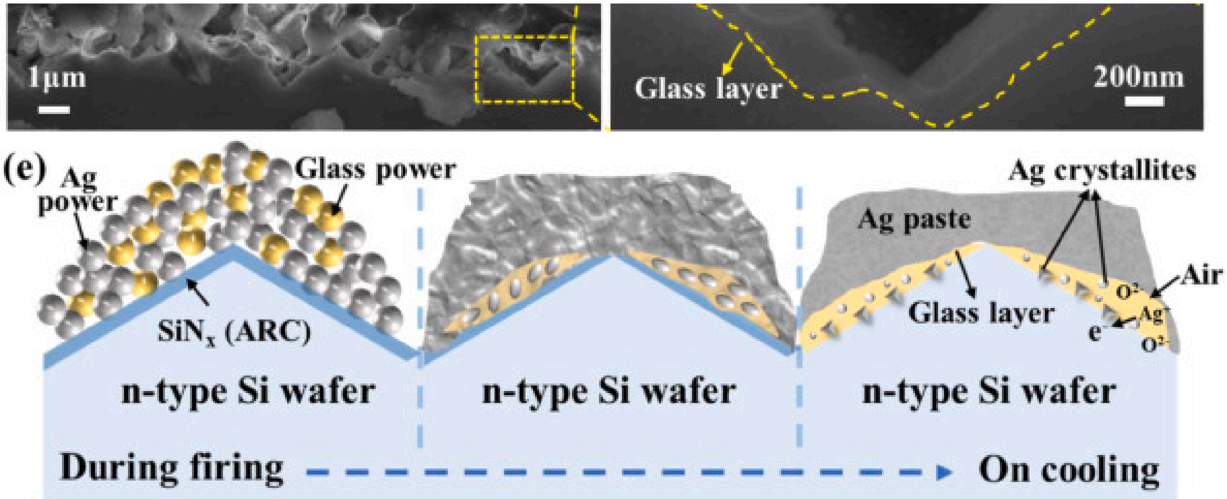


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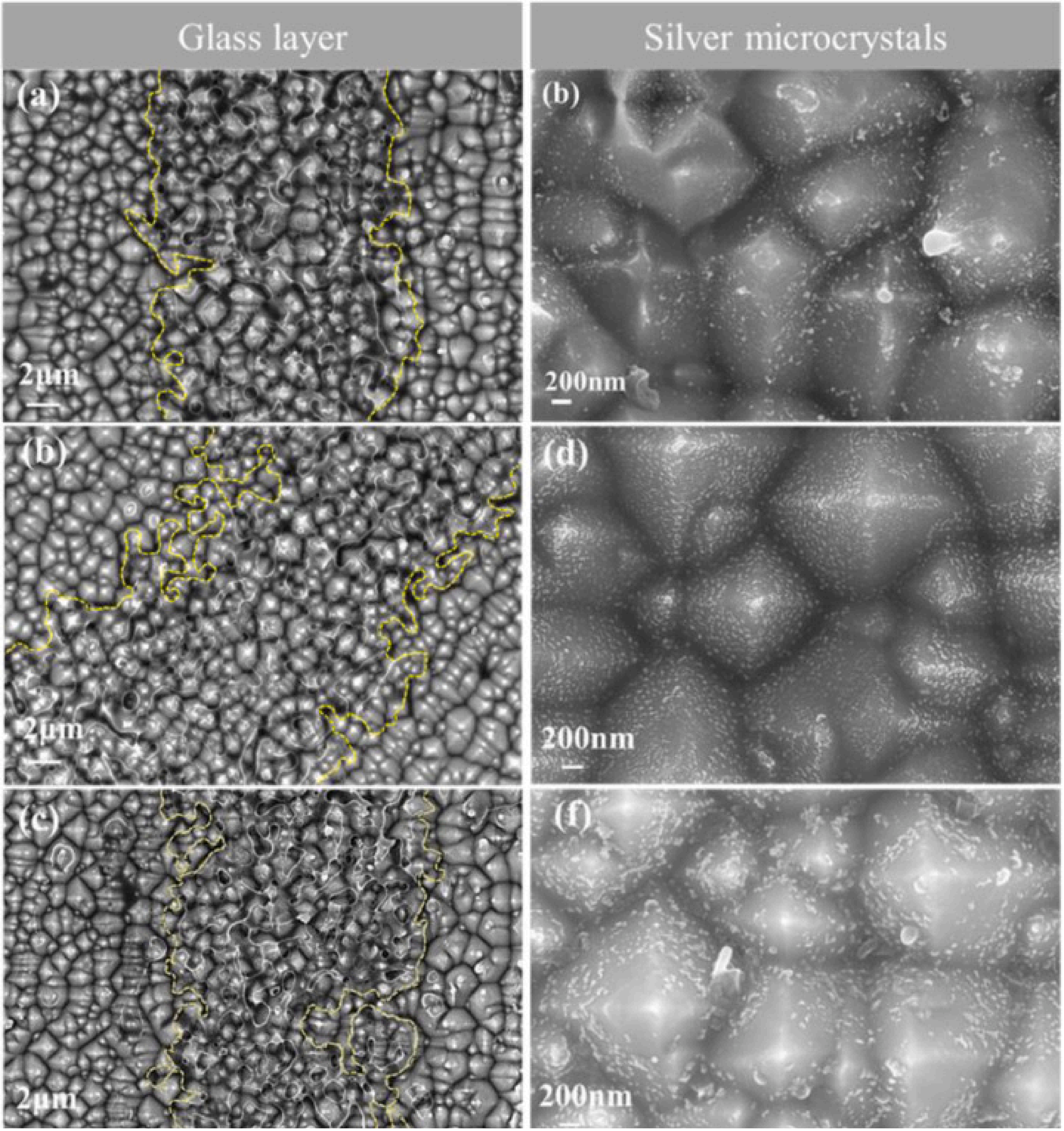
Fig. 7. The SEM image of the cross-section of the contact interface between the silver electrode and the silicon substrate: (a) LC-1, (b) LC-2, (c) LC-3, (d) LC-4, (e) Schematic diagram of the glass frit forming a contact between the silver electrode and the silicon wafer. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

The SEM image presented the cross-section of the contact interface between the silver electrode and the silicon substrate in Fig. 7(a–d). It was evident that there were fewer silver particles in the interface of LC-1 cell, the L1 glass frits with high phase transition temperature melts later during the sintering process. It exhibited poor wettability on the silicon substrate and limited solubility for silver powder, resulting in low-density silver grids. In contrast, the Ag-Si interfaces in the other three cells displayed silver microcrystals of varying sizes. L4 glass frit has a low softening point, causing it to melt prematurely. This can lead to excessive flow and the formation of a non-uniform glass layer on the silicon substrate. Therefore, some areas of the cell may experience poor current collection, which can affect the overall electrical performance of the cell. This can also lead to over-etching of the silicon emitter, damaging the pn junction and allowing impurities to diffuse into the cell through the interfacial glass layer. These effects increase shunt resistance and carrier recombination losses, ultimately reducing cell efficiency. The glass frits employed in LC-2 and LC-3 cells are characterized by suitable high-temperature wettability. The condition foster the generation of nanoscale silver grains on the silicon substrate, which improves ohmic contact after LECO, ultimately improved the electrical performance of the solar cells.

The silver grids were etched using a wet method, immersing the cells in aqua regia ($HCl: HNO_3=3:1$) and storing the samples in a dark room for 24h. Excess acid on the cells was then rinsed away multiple times with deionized water and ethanol. The morphology of the glass layer between the grid lines and the silicon substrate was observed, as shown in Fig. 8(a)–(c), (e), and (g). After soaking in HF to remove the glass layer, silver microcrystal distribution on the emitter surface was observed, as shown in Fig. 8(b)–(d), (f), and (h).


Glass layer
Silver microcrystals
(a)
2μm
(b)
200nm
(b)
2μm
(d)
200nm
(c)
2μm
(f)
200nm

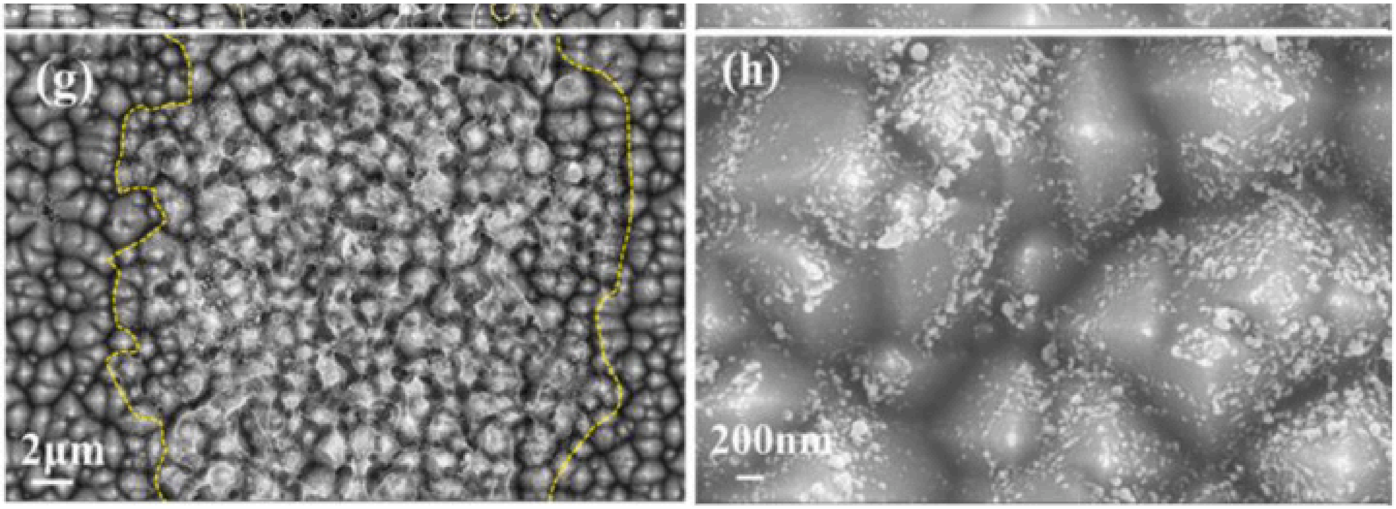


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Fig. 8. Surface morphology of solar cell grids after aqua regia immersion and surface morphology of silicon substrate after hydrofluoric acid etching. (a) and (b): LC-1cell; (c) and (d): LC-2cell; (e) and (f): LC-3cell; (g) and (h): LC-4cell. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

The glass layers for each sample group were delineated with yellow dashed lines. Notable differences in the glass layers corresponding to the various glass frits were observed. The cell of LC-1 exhibited a discontinuous glass layer in the center, which can impede electron transport and ultimately reduce the efficiency of the solar cell. In contrast, the glass layers of the LC-2 and LC-3 cells were relatively uniform and exhibit better continuity. The contact angle test of the contour morphology of the molten glass can tell that the glass frit used in the LC-3 cell has better wettability, allowing the glass layer to spread more evenly towards the edges of the grid lines. Suitable content of TeO_2 and Bi_2O_3 in L2 and L3 glass frits enhance the adhesion and diffusion of silver particles at the silicon surface. These oxides improve the fluidity of the glass, enabling it to better wet the silver particles and the silver-silicon interface during high-temperature sintering [46,47]. Thereby uniformly coating the surface and preventing localized thickening or unevenness of the glass layer. In contrast, the LC-4 cell features non-uniform glass layer, likely due to the excessive content of Bi_2O_3 in the L4 glass frit, which enhances its ability to melt silver. This resulted in the accumulation of silver particles within the glass layer during the sintering process. Consequently, after cooling, the increased unevenness of the glass layer lengthens the electron transport path, which increases the migration distance. This reduces the efficiency of electron transport and leads to greater power loss.

It is well established that glass undergoes melting and liquefaction during high-temperature sintering, allowing it to wet silver particles and etch the SiN_x layer. Silver dissolved in the glass can precipitate as microcrystals or silver nanoparticles on the surface of the crystalline silicon emitter. There were few silver particles in the silicon substrate of LC-1 cell. Conversely, the LC-2 cell features a uniform and dense distribution of silver nanoparticles, with small pits at the tips of the pyramids marking the locations where silver microcrystals were removed from the glass layer. The base of the pyramids is densely populated with silver nanoparticles. The silver microcrystals of the silicon substrate of the LC-3 cell are uniformly distributed around the bottom of the tower. The surface of the silicon substrate in the LC-4 cell exhibits a noticeable distribution of unevenly aggregated silver particles. In LC-4 cell, the high Bi_2O_3 content in the L4 glass frit lowers its softening point, causing premature melting. This results in silver particles aggregating unevenly on the silicon substrate surface. Excessive accumulation of silver particles in the glass phase or an overly thick silver layer can increase surface recombination, leading to a sharp rise in leakage current. This, in turn, reduces the V_{oc} and FF of the solar cell. [48,49]. Additionally, during the sintering process, this can cause insufficient densification of the silver paste layer, affecting the width and uniformity of the metal grids. As a result, current losses increase when passing through these regions. Moreover, excessive growth of silver dendrites can damage the emitter, further compromising the photovoltaic efficiency of the solar cell.

The performance parameters of cells fabricated with four types of glass frits are detailed in Table S2. For each batch of glass frit formulation, ten solar cell specimens will be fabricated and subsequently characterized for their electrical performance parameters, with the measured data subjected to statistical averaging values. Based on the table analysis, it is clear that the LC-4cell has the lowest efficiency. The main reason is the high content of Bi_2O_3 and TeO_2 in the glass, which causes it to melt and flow too early. This increases the flowability of the paste, improving contact and reducing series resistance. While this improves the flowability of the paste, enhances contact, and reduces series resistance. It also leads to unevenness at the silicon substrate-metal contact interface during sintering, which affects the distribution of silver crystallites. This leads to increased recombination of charge carriers at the contact interface, which negatively impacts other electrical parameters. In the LC-1cell, the absence of Bi_2O_3 results in poor glass flowability and wettability. The uneven contact leads to inefficient current collection, which in turn affects the FF and overall efficiency of the solar cell. The average efficiency difference between the LC-2 and LC-3cells is identical, but their electrical parameters differ. The glass frit used in LC-2cells has a higher content of TeO_2. TeO_2 has good light transmittance, which reduces light reflection losses. This allows more photons to generate additional photogenerated charge carriers, thereby increasing the short-circuit current (I_{sc}). At the same time, the higher R_{sh} significantly reduces reverse current leakage. However, the higher TeO_2 content also increases the viscosity of the glass at high temperatures. This affects the uniform distribution of the silver paste, reduces contact quality, and increases R_s. On the other hand, the low R_s of LC-3cell indicates better Ag-Si contact. The L3 glass frit contains an appropriate amount of bismuth oxide and tellurium oxide, which gives the glass frit a suitable glass transition temperature and good wettability. During the sintering process, it can fully wet the silver powder and etch the SiN_x layer,and form a uniform glass layer after cooling. This results in uniformly dense nano-silver crystallites in the silicon substrate. After LECO treatment, a good ohmic contact is formed between the metalized layer and the P^+ emitter, reducing contact resistance. This leads to a smaller V_{oc} decrease and a significant increase in FF, which is important for improving the overall efficiency of the cell.

Fig. 9(a)–(d) present three-dimensional morphological images obtained using a 3D microscope after sintering the silver grid of different solar cells. Ten parallel aspect ratio tests were conducted randomly on the silver grid of various solar cells, and the differences in height and width were within the measurement uncertainty range. The data on grid line height, width, and aspect ratio are shown in Fig. 9(e)–(g). The LC-3 solar cell demonstrated the optimal aspect ratio for its grid lines. Under identical conditions, a high aspect ratio in the silver electrodes facilitates effective current collection between the grid lines and enhances conduction to the external circuit [50], ultimately improving the photovoltaic conversion efficiency of the cell. Compared to standard production line cells, LC-3cell fill factor is significantly increased.

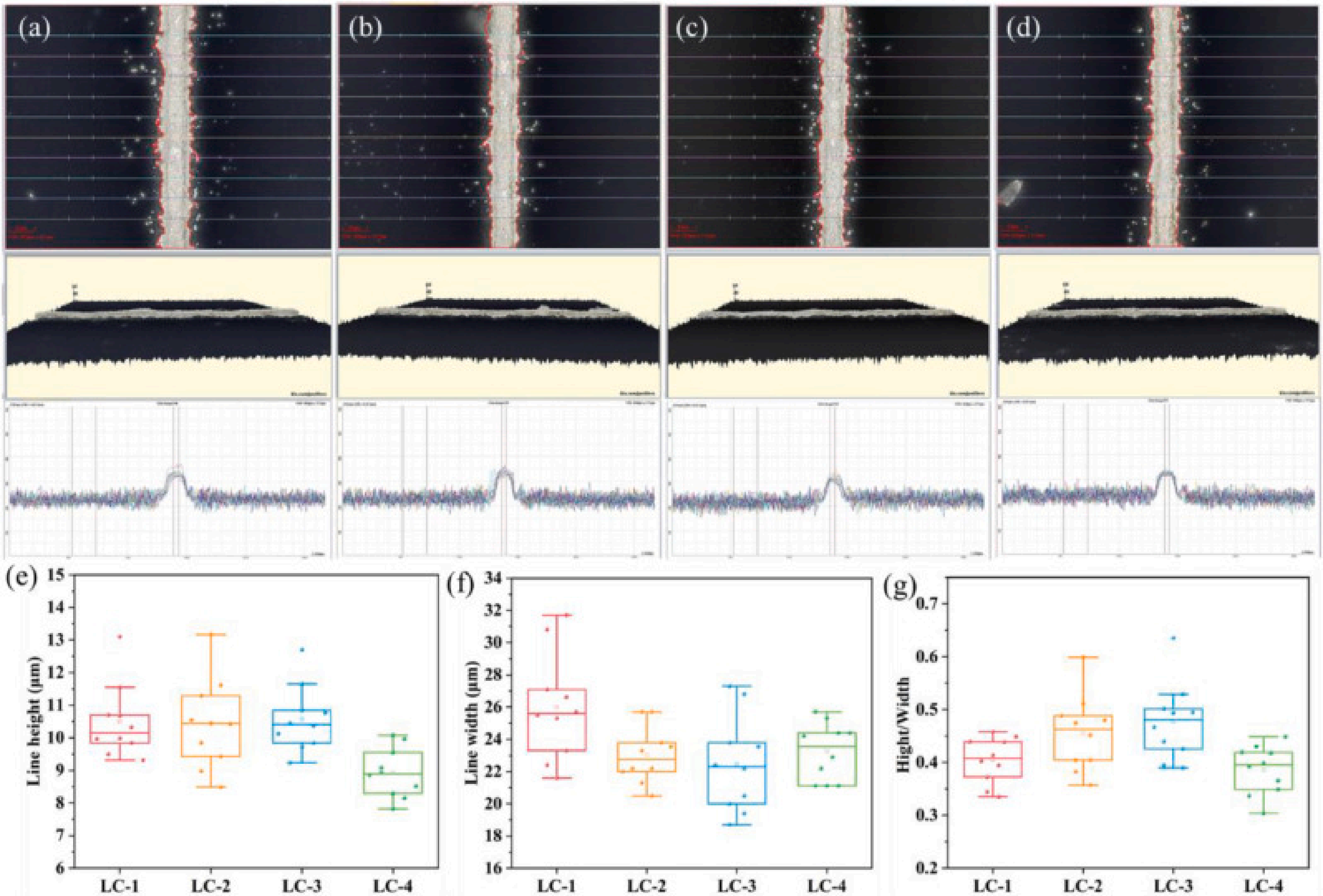

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Fig. 9. 3D microscope images of sintered silver grids of different cells (a)LC-1, (b) LC-2, (c) LC-3, (d) LC-4; (e) The height of the silver grids; (f) The width of the silver finger; (g) The height/width ratio of the silver grids.

4. Conclusion

This research reveals that altering the Bi_2O_3 and TeO_2 content in low-lead glass increases non-bridging oxygens (NBOs), disrupting the network structure. As a result, the viscosity and softening temperature decrease, allowing the glass to flow and spread more rapidly at lower temperatures. Good wettability of the glass frit promotes the formation of dense silver grid lines with excellent aspect ratios. After chemically etching away the silver grids, the glass in LC-2 and LC-3 cells was found to be evenly dispersed over the n^+ emitter. This uniformity facilitated a stable and consistent connection between the silver grids and the silicon substrate. Further etching the glass layer, a large number of nano-silver particles deposited on the silicon emitter surface. These particles provide many contact points, forming a high-quality interface between silver and silicon, which reduces carrier recombination. Unsuitable wettability can cause silver particle aggregation within the glass layer or excessive spreading of the glass frit, leading to over-etching of the silicon surface. LC-4 exhibited the lowest R_s compared to the cells from the standard production line. This was attributed to the aggregation of silver crystallites on the silicon substrate, forming larger contact points. However, such "over-contact" can enhance surface recombination, leading to higher recombination currents and reduced V_{oc}. Notably, the LC-3 cell achieved a significant boost in the fill factor while maintaining satisfactory contact. This highlights the importance of a stable and low-resistance interface for effectively collected charge carriers. These findings demonstrate that glass frits with good wettability facilitate the formation of uniform, continuous, and dense metal electrodes, enabling high-quality interfacial contact between the metal and silicon substrate.

CRediT authorship contribution statement

Qian Li: Writing – original draft, Investigation, Data curation. **Wenbin Sun:** Validation, Methodology. **Yinghu Sun:** Validation, Methodology. **Xiaojie Liu:** Methodology, Conceptualization. **Hanying Wang:** Methodology, Investigation, Funding acquisition. **Shenghua Ma:** Writing – review & editing, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

The following is the Supplementary data to this article.

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Multimedia component 1.

Recommended articles

Data availability

Data will be made available on request.

References

[1] C. Battaglia, A. Cuevas, S. De Wolf
High-efficiency crystalline silicon solar cells: status and perspectives
Energy Environ. Sci., 9 (2016), pp. 1552-1576, 10.1039/c5ee03380b ↗
View in Scopus ↗ Google Scholar ↗

[2] P.-J. Ribeyron
Crystalline silicon solar cells: better than ever
Nat. Energy, 2 (2017), pp. 1-2, 10.1038/nenergy.2017.67 ↗
Google Scholar ↗

[3] H. Ullah, S. Czapp, S. Szultka, H. Tariq, U.B. Qasim, H. Imran

Crystalline silicon (c-Si)-based tunnel oxide passivated contact (TOPCon) solar cells: a review
Energies, 16 (2023), p. 715, 10.3390/en16020715 ↗
View in Scopus ↗ Google Scholar ↗

[4] Q. Wang, H. Peng, S. Gu, K. Guo, W. Wu, B. Li, L. Li, N. Yuan, J. Ding
High-efficiency n-TOPCon bifacial solar cells with selective poly-Si based passivating contacts
Sol. Energy Mater. Sol. Cells, 259 (2023), Article 112458, 10.1016/j.solmat.2023.112458 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[5] X. Li, Q. Wang, X. Dong, J. Li, X. Zhang, N. Yuan, L. Li, J. Ding
High-efficiency n-TOPCon cells ensured by an emitter preparation process without post-oxidation
Sol. Energy Mater. Sol. Cells, 279 (2025), Article 113223, 10.1016/j.solmat.2024.113223 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[6] D.K. Ghosh, S. Bose, G. Das, S. Acharyya, A. Nandi, S. Mukhopadhyay, A. Sengupta
Fundamentals, present status and future perspective of TOPCon solar cells: a comprehensive review
Surf. Interfaces., 30 (2022), Article 101917, 10.1016/j.surfin.2022.101917 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[7] B. Kafle, B.S. Goraya, S. Mack, F. Feldmann, S. Nold, J. Rentsch
TOPCon – technology options for cost efficient industrial manufacturing
Sol. Energy Mater. Sol. Cells, 227 (2021), Article 111100, 10.1016/j.solmat.2021.111100 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[8] Y. Zeng, H. Tong, C. Quan, L. Cai, Z. Yang, K. Chen, Z. Yuan, C.-H. Wu, B. Yan, P. Gao, J. Ye
Theoretical exploration towards high-efficiency tunnel oxide passivated carrier-selective contacts (TOPCon) solar cells
Sol. Energy, 155 (2017), pp. 654-660, 10.1016/j.solener.2017.07.014 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[9] Z. Zhang, Y. Zeng, C.-S. Jiang, Y. Huang, M. Liao, H. Tong, M. Al-Jassim, P. Gao, C. Shou, X. Zhou, B. Yan, J. Ye

Carrier transport through the ultrathin silicon-oxide layer in tunnel oxide passivated contact (TOPCon) c-Si solar cells
Sol. Energy Mater. Sol. Cells, 187 (2018), pp. 113-122, 10.1016/j.solmat.2018.07.025 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[10] Y. Li, R. Zhou, Z. Chen, Y. Li, X. Cheng, B. Zhang, J. Chen, Y. Lin, F. Pan
Insight into the contact mechanism of Ag/Al-Si Interface for the Front-side metallization of TOPCon silicon solar cells
Small Methods (2024), Article 2400707, 10.1002/smtd.202400707 ↗
Google Scholar ↗

[11] X. Sun, J. Xing, Y. Yang, X. Yuan, H. Li, H. Tong
Ohimc contact formation mechanism of silver-aluminum paste metallization on the p^+ emitter of n-Type crystalline silicon solar cells
J. Electron. Mater., 51 (2022), pp. 5717-5722, 10.1007/s11664-022-09821-2 ↗
View in Scopus ↗ Google Scholar ↗

[12] M.S. Jeong, S.H. Lee, S. Choi, K.H. Min, J.H. Lee, M.G. Kang, K.T. Jeong, S.K. Kwak, H.-e. Song, T.K. Lee, S. Park
Origin of screen-printed metal contact losses in crystalline silicon solar cells
ACS Appl. Energy Mater., 6 (2023), pp. 11983-11992, 10.1021/acsaem.3c02143 ↗
View in Scopus ↗ Google Scholar ↗

[13] J. Zhang, H. Tong, X. Sun, G. Li, H. Li, Y. Yang, X. Yuan, C. Liu, H. Li
An investigation on determinants of silver paste metallization contact performance on crystalline silicon solar cells
J. Mater. Sci. Mater. Electron., 31 (2020), pp. 5752-5759, 10.1007/s10854-020-03145-9 ↗
View in Scopus ↗ Google Scholar ↗

[14] M. Fırat, H. Sivaramakrishnan Radhakrishnan, S. Singh, F. Duerinckx, M. Recamán Payo, L. Tous, J. Poortmans

Industrial metallization of fired passivating contacts for n-type tunnel oxide passivated contact (n-TOPCon) solar cells
Sol. Energy Mater. Sol. Cells, 240 (2022), Article 111692, 10.1016/j.solmat.2022.111692 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[15] Y. Fan, S. Zou, Y. Zeng, L. Dai, Z. Wang, Z. Lu, H. Sun, X. Zhou, B. Liao, X. Su
Investigation of the Ag-Si contact characteristics of boron emitters for n-tunnel oxide-passivated contact solar cells metallized by laser-assisted current injection treatment
Sol. RRL, 8 (2024), Article 2400268, 10.1002/solr.202400268 ↗
View in Scopus ↗ Google Scholar ↗

[16] S. Groser, E. Krassowski, S. Swatek, H. Zhao, C. Hagendorf
Microscale contact formation by laser enhanced contact optimization
IEEE J. Photovoltaics, 12 (2022), pp. 26-30, 10.1109/jphotov.2021.3129362 ↗
View in Scopus ↗ Google Scholar ↗

[17] T. Guo, Y. Sun, Y. Song, S. Ma
Effect of MoO_3/WO_3 modulation on high-temperature wettability and crystallization behavior of glass and its application in solar cell
Ceram. Int., 50 (2024), pp. 12566-12579, 10.1016/j.ceramint.2024.01.166 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[18] C. Yang, J. Bai, G. Wang, H. Wang, S. Ma
Effect of Na_2O content on wettability, crystallization and performances of sealing glass
J. Mater. Res. Technol., 23 (2023), pp. 4117-4134, 10.1016/j.jmrt.2023.02.073 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[19] H. Li, X. Sun, J. Xing, Y. Yang, X. Yuan, H. Tong
Effect of glass frit composition on reliability of silver paste metallization in crystalline silicon solar cells
Mater. Express., 11 (2024), Article 056303, 10.1088/2053-1591/ad48e4 ↗
View in Scopus ↗ Google Scholar ↗

[20] H. Sun, X. Yu, S. Quan, W. Li, Z. Qian, J. Li, G. Gan
Effect of lead-free glass on the current transmission method at the Ag/Si interface in crystalline silicon solar cells
RSC Adv., 14 (2024), pp. 6048-6057, 10.1039/d4ra00456f ↗
View article View in Scopus ↗ Google Scholar ↗

[21] J. Chen, Y. Li, W. Miao, Q. Lei, M. Li
Dependence of glass transition on the structure in Bi-B-Zn oxide glass
J. Alloys Compd., 742 (2018), pp. 151-158, 10.1016/j.jallcom.2018.01.188 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[22] Y.B. Saddeek, K.A. Aly, K.S. Shaaban, A.M. Ali, M.M. Alqhtani, A.M. Alshehri, M.A. Sayed, E.A. Abdel Wahab
Physical properties of B_2O_3-TeO_2-Bi_2O_3 glass system
J. Non-Cryst. Solids, 498 (2018), pp. 82-88, 10.1016/j.jnoncrysol.2018.06.002 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[23] A. Madhu, C.M. Joseph, S.T. Dadami, N. Srinatha
Structural and spectroscopic studies of V_2O_5 incorporated lithium-bismuth-boro-tellurite glass ceramics
Mater. Today Proc., 62 (2022), pp. 5489-5493, 10.1016/j.matpr.2022.04.154 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[24] K. Vamshi Babu, N. Chanakya, S. Aravind, G. Upender
Structural, Thermal, Optical and DC conductivity studies of $30TeO_2$-$40B_2O_3$-$(30-x)BaO$-xNa_2O ($0 \le x \le 30$ mol%) glass system
J. Inorg. Organomet. Polym. Mater., 34 (2023), pp. 1738-1750, 10.1007/s10904-023-02914-3 ↗
Google Scholar ↗

[25] B. Zhou, Q. Ma, C. Cui, S. Ma, J. Bai, H. Wang
Preparation of the lead-free graphene-glass frit composites for crystalline silicon solar cells and investigation of performance
Sol. Energy, 211 (2020), pp. 988-998, 10.1016/j.solener.2020.10.044 ↗

View PDF View article View in Scopus ↗ Google Scholar ↗

[26] M. Anand Pandarinath, P. Muralikrishna, D. Suresh Babu, B. Vijaya Kumar, G. Upender
Vibrational, thermal and optical studies of $30TeO_2$-$39.5B_2O_3$-(30-x)ZnO-xLi_2O-$0.5V_2O_5$ ($0 \leq x \leq 30$ mol%) glass system
J. Non-Cryst. Solids, 566 (2021), Article 120875, 10.1016/j.jnoncrysol.2021.120875 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[27] A.M. Efa, K.A. Matori, M.H.M. Zaid, C.A.C. Abdullah, N. Zainuddin, M.Z.H. Mayzan, S.L. Kamis
Effect of B_2O_3 on the physical and mechanical properties of calcium fluoroaluminosilicate glass system
Silicon, 16 (2024), pp. 5327-5336, 10.1007/s12633-024-03072-y ↗
View in Scopus ↗ Google Scholar ↗

[28] L. Han, J. Song, Q. Zhang, Z. Luo, A. Lu
Crystallization, structure and characterization of MgO-Al_2O_3-SiO_2-P_2O_5 transparent glass-ceramics with high crystallinity
J. Non-Cryst. Solids, 481 (2018), pp. 123-131, 10.1016/j.jnoncrysol.2017.10.028 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[29] K. Jha, M. Jayasimhadri
Spectroscopic investigation on thermally stable Dy^{3+} doped zinc phosphate glasses for white light emitting diodes
J. Alloys Compd., 688 (2016), pp. 833-840, 10.1016/j.jallcom.2016.07.024 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[30] Y. Li, Q. Zhang, J. Song, X. Gao, W. Tang, A. Lu
Glass formation and spectral studies of PbO and Bi_2O_3 modified TeO_2-ZnO glasses
J. Non-Cryst. Solids, 483 (2018), pp. 43-49, 10.1016/j.jnoncrysol.2017.12.053 ↗
View PDF View article Google Scholar ↗

[31] J. Liu, X. Xu, T. Zheng, Y. Guo, J. Lv

Effect of Bi_2O_3 content on the structure and properties of Bi_2O_3-B_2O_3-BaO-ZnO glass
J. Non-Cryst. Solids, 575 (2022), Article 121211, 10.1016/j.jnoncrysol.2021.121211 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[32] S. Kurumada, R. Yamanashi, K. Sugita, K. Kubota, H. Ito, S. Ikemoto, C. Chen, T. Moriyama, S. Muratsugu, M. Tada, T. Koitaya, T. Ozaki, M. Yamashita
Mechanochemical synthesis of Non-solvated dialkylalumanyl anion and XPS characterization of Al(I) and Al(II) species
Chem. Eur J., 30 (2023), Article e202303073, 10.1002/chem.202303073 ↗
Google Scholar ↗

[33] X. Lin, Z. Wang, X. Jiang, T. Ning, Y. Jiang, A. Lu
Effect of Al_2O_3/SiO_2 mass ratio on the structure and properties of medical neutral boroaluminosilicate glass based on XPS and infrared analysis
Ceram. Int., 49 (2023), pp. 38499-38508, 10.1016/j.ceramint.2023.09.180 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[34] K. Yliniemi, D. Wragg, B.P. Wilson, H.N. McMurray, D.A. Worsley, P. Schmuki, K. Kontturi
Formation of Pt/Pb nanoparticles by electrodeposition and redox replacement cycles on fluorine doped tin oxide glass
Electrochim. Acta, 88 (2013), pp. 278-286, 10.1016/j.electacta.2012.10.089 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[35] C.M. Smyth, G. Zhou, A.T. Barton, R.M. Wallace, C.L. Hinkle
Controlling the Pd metal contact polarity to trigonal Tellurium by atomic Hydrogen-removal of the native Tellurium oxide
Adv. Mater. Interfac., 8 (2021), Article 2002050, 10.1002/admi.202002050 ↗
View in Scopus ↗ Google Scholar ↗

[36] Y. Fang, S. Meng, J. Hou, Y. Liu, Y. Guo, G. Zhao, J. Zou, L. Hu
Experimental study of growth of silver nanoparticles embedded in Bi_2O_3-SiO_2-B_2O_3 glass

J. Alloys Compd., 809 (2019), Article 151725, 10.1016/j.jallcom.2019.151725 ↗

View PDF View article View in Scopus ↗ Google Scholar ↗

[37] N.A. Zhuk, B.A. Makeev, A.V. Koroleva, O.V. Petrova, S.V. Nekipelov
Synthesis, XPS and NEXAFS spectroscopy study of Zn, Cr codoped bismuth tantalate pyrochlores
Ceram. Int., 50 (2024), pp. 21342-21347, 10.1016/j.ceramint.2024.03.245 ↗

View PDF View article View in Scopus ↗ Google Scholar ↗

[38] Y. Liu, J. Chen, G. Zheng
Study on the wetting mechanism between Hot-melt nano glass powder and different substrates
Micromachines, 13 (2022), p. 1683, 10.3390/mi13101683 ↗

View in Scopus ↗ Google Scholar ↗

[39] N. Sobczak, M. Singh, R. Asthana
High-temperature wettability measurements in metal/ceramic systems-Some methodological issues
Solid State Mater. Sci., 9 (2005), pp. 241-253, 10.1016/j.cossms.2006.07.007 ↗

View PDF View article View in Scopus ↗ Google Scholar ↗

[40] L. He, Z. Luo, H. Liang, N. Ma, X. Liu, X. Wang, Z. Zhou, A. Lu
Structure, properties and wetting behaviour on Si substrates of lead-free low melting-temperature Li_2O-ZnO-B_2O_3-WO_3 glasses
Ceram. Int., 50 (2024), pp. 24451-24460, 10.1016/j.ceramint.2024.04.178 ↗

View PDF View article View in Scopus ↗ Google Scholar ↗

[41] S. Zhao, D. Xing, K. Zhen, Z. Niu, G. Li
Wetting and spreading behavior of Bi_2O_3-B_2O_3 (-ZnO) glass on silver
J. Mater. Eng. Perform., 32 (2023), pp. 10626-10636, 10.1007/s11665-023-07902-9 ↗

View in Scopus ↗ Google Scholar ↗

[42] Y. Kim, J. Huh, H. Kim
Effect of basicity of glass frits on electrical properties of Si solar cells

Sol. Energy Mater. Sol. Cells, 185 (2018), pp. 97-103, 10.1016/j.solmat.2018.05.015 ↗

View PDF View article View in Scopus ↗ Google Scholar ↗

[43] C. Terrados, D. González-Francés, V. Alonso, M.A. González, J. Jiménez, O. Martínez
Comparison of outdoor and indoor PL and EL images in Si solar cells and panels for defect detection and classification
J. Electron. Mater., 52 (2023), pp. 5189-5198, 10.1007/s11664-023-10535-2 ↗

View in Scopus ↗ Google Scholar ↗

[44] Y.-S. Lai, S.-S. Lai, Y.-J. Li, H.-J. Lin, T.-H. Chiang
Investigation of SiO_2-B_2O_3-ZnO-Bi_2O_3 glass frits on the interface reaction of silver front contacts
J. Alloys Compd., 858 (2021), Article 157646, 10.1016/j.jallcom.2020.157646 ↗

View PDF View article View in Scopus ↗ Google Scholar ↗

[45] X. Sun, S. Yao, J. Xing, J. Zhang, Y. Yang, H. Li, H. Tong, X. Yuan
Mechanism of silver/glass interaction in the metallization of crystalline silicon solar cells
Mater. Res. Express, 7 (2020), Article 016315, 10.1088/2053-1591/ab6ca2 ↗

View in Scopus ↗ Google Scholar ↗

[46] Y. Kim, T. Nakayama, H. Kim
Effect of Te-based glass on contact formation and electrical properties in Si solar cells
J. Alloys Compd., 829 (2020), 10.1016/j.jallcom.2020.154500 ↗

Google Scholar ↗

[47] Y. Sun, H. Xue, C. Yang, Y. Tian, S. Ning, S. Ma, H. Wang
Improvement of wettability of Te-modified lead-free glass frit and its effect to front side silver paste in crystalline silicon solar cells
Sol. Energy Mater. Sol. Cells, 253 (2023), 10.1016/j.solmat.2023.112214 ↗

Google Scholar ↗

[48] C. Kim, J.W. Choi, S. Choi, S. Kim, H. Park, H.E. Song, S.S. Yoon, J.Y. Huh, Y. Kang, H.S. Lee, D. Kim

Effects of Current-injection firing with Ag paste in a boron emitter

Sci. Rep., 6 (2016), Article 21553, 10.1038/srep21553 ↗

View in Scopus ↗ Google Scholar ↗

[49] W. Li, T. Wu, R. Jiao, B.-P. Zhang, S. Li, Y. Zhou, L. Li

Effects of silver nanoparticles on the firing behavior of silver paste on crystalline silicon solar cells

Colloids Surf., 466 (2015), pp. 132-137, 10.1016/j.colsurfa.2014.11.018 ↗

View PDF View article View in Scopus ↗ Google Scholar ↗

[50] Y. Jang, I.H. Tambunan, H. Tak, V.D. Nguyen, T. Kang, D. Byun

Non-contact printing of high aspect ratio Ag electrodes for polycrystalline silicone solar cell with electrohydrodynamic jet printing

Appl. Phys. Lett., 102 (2013), Article 123901, 10.1063/1.4798332 ↗

View in Scopus ↗ Google Scholar ↗

Cited by (6)

Based on silver/glass frit screening & AHP-EWM evaluation: Performance and production verification of TOPCon/LECO photovoltaic fine-grid silver pastes

2026, Solar Energy Materials and Solar Cells

Show abstract

Comprehensive review of photovoltaic paste: Materials, processing, and performance optimization

2026, Solar Energy Materials and Solar Cells

Show abstract

Effects of free oxygen concentration in glass structures with different B/Ga ratios on the metallization of p+ emitter during laser-enhanced contact optimization

2025, Solar Energy Materials and Solar Cells

Show abstract

Study on the sintering behavior, microstructure, and properties of RO-TiO2-B2O3-SiO2 glass-ceramics applied to silver paste LTCC

2025, Journal of Alloys and Compounds

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